DEEP LEARNING FINAL PROJECT

Finding the Right Law, in Khmer

Can a model match everyday Khmer questions with the correct legal articles?

EXAMPLE USER QUESTION

តើកិច្ចសន្យាបង្កើតឡើងដោយរបៀបណា?

1,976	4	200
LEGAL ARTICLES	NEURAL APPROACHES	NATURAL TEST QUESTIONS

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A keyword match is not always a meaning match

តើកិច្ចសន្យាបង្កើតឡើងដោយរបៀបណា?

BM25 / KEYWORD SEARCH

Finds overlapping words

One recorded top match was Article 623 — about dividing proceeds, not forming a contract.

NEURAL SEMANTIC RETRIEVAL

Compares meaning

A trained encoder maps questions and articles into vectors and ranks them by similarity.

Goal: retrieve the relevant article, not generate a new law.

First, we had to make the Khmer PDFs readable

RAW EXTRACTED TEXT

maRta 336>- karbegáltkic©snña...

AFTER LIMON → UNICODE CONVERSION

មាត្រា ៣៣៦.- ការបង្កើតកិច្ចសន្យា...

Official PDFs
Source documents

→

Convert Limon
Legacy font mapping

→

Normalize Khmer
Character ordering

→

Split into articles
Preserve metadata

1,304 Civil Code articles

672 Criminal Code articles

We had legal articles, but not enough question–answer pairs

WEAK SUPERVISION: A TRAINING PAIR

Article title → Article body

We removed titles from article bodies to avoid trivial text matching.

1,176	147	148

Separate main benchmark: 200 natural Khmer questions · 261 reference articles excluded from training.

Four ways to teach a model to retrieve legal articles

A1 · START FROM SCRATCH

BiLSTM

Khmer word tokens

↓

1-layer BiLSTM

↓

512-d vector

1.97M trainable

A2 · FREEZE THE BACKBONE

XLM-R probe

Pretrained XLM-R

↓

Frozen backbone

↓

Linear head

0.59M trainable

A3 · ADAPT ALL LAYERS

XLM-R fine-tune

Pretrained XLM-R

↓

12 transformer layers

↓

768-d vector

278.04M trainable

A4 · KHMER PRETRAINING

PrahokBART

Khmer pretrained BART

↓

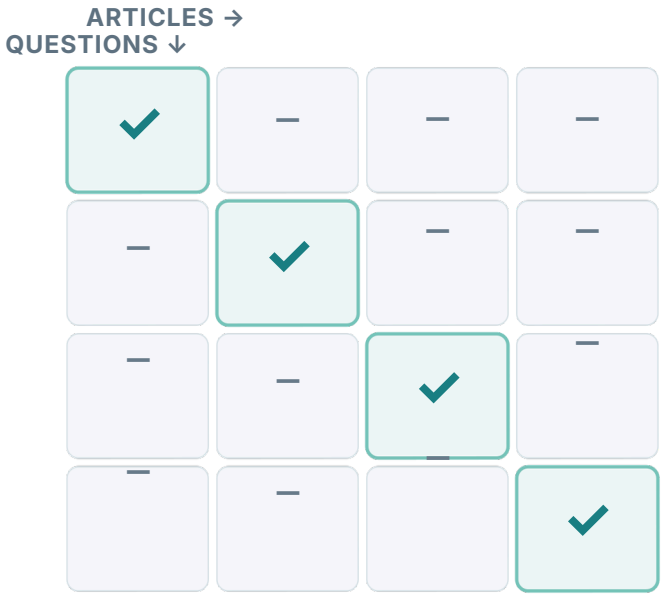
Encoder + pooling

↓

512-d vector

35.83M trainable

Contrastive learning: bring matching pairs closer



One correct pair per row

Otherarticlesinthebatchactasnegative examples.

A3 batch size: 16 · 15 in-batch negatives

Similarity → InfoNCE loss (τ = 0.05) → Backpropagation → Updated model weights

EXPERIMENTAL EVIDENCE

Experimental setup & compute

Same dataset and evaluation protocol. Different learning strategies.

FOUR TRAINED APPROACHES

APPROACH	TRAINABLE	HARDWARE
A1 · BiLSTM	1.97M	CPU
A2 · Linear probe	0.59M	CPU
A3 · Full fine-tune	**278.04M**	**T4 GPU**
A4 · PrahokBART	35.83M	CPU (14 threads)

A3 HYPERPARAMETER SEARCH

6 training runs

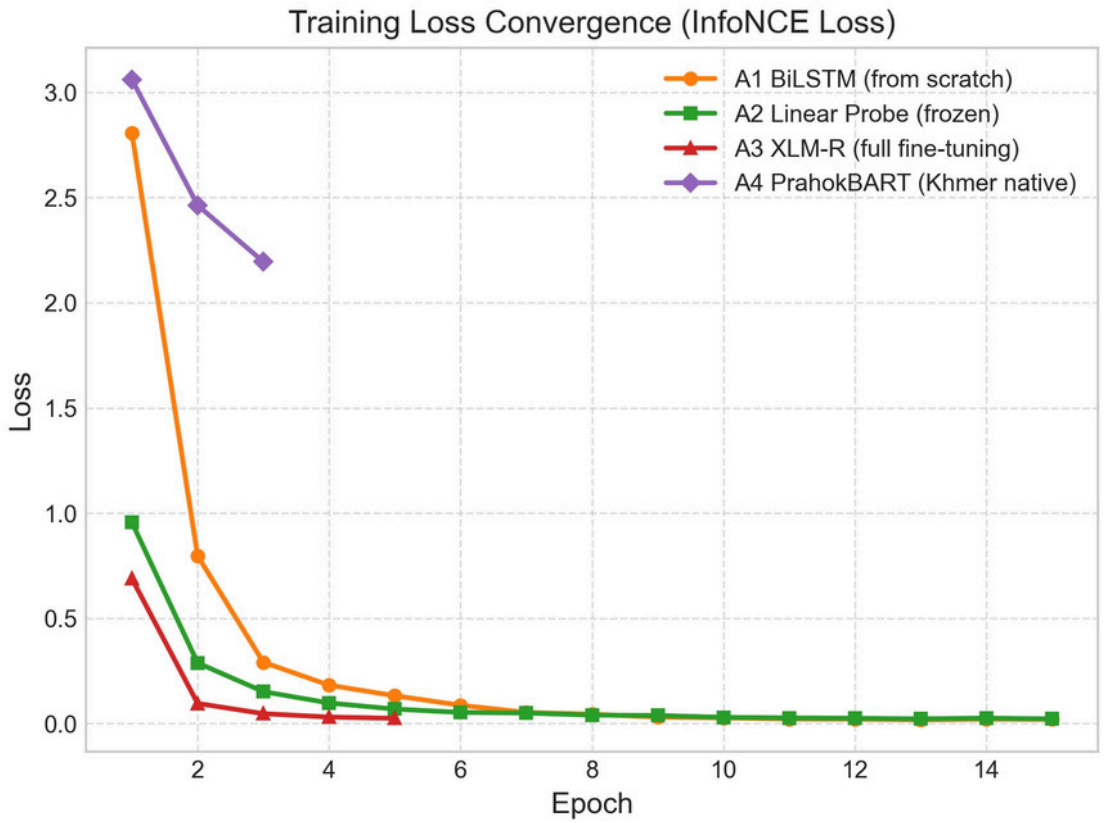
3 learning rates × 2 weight decays

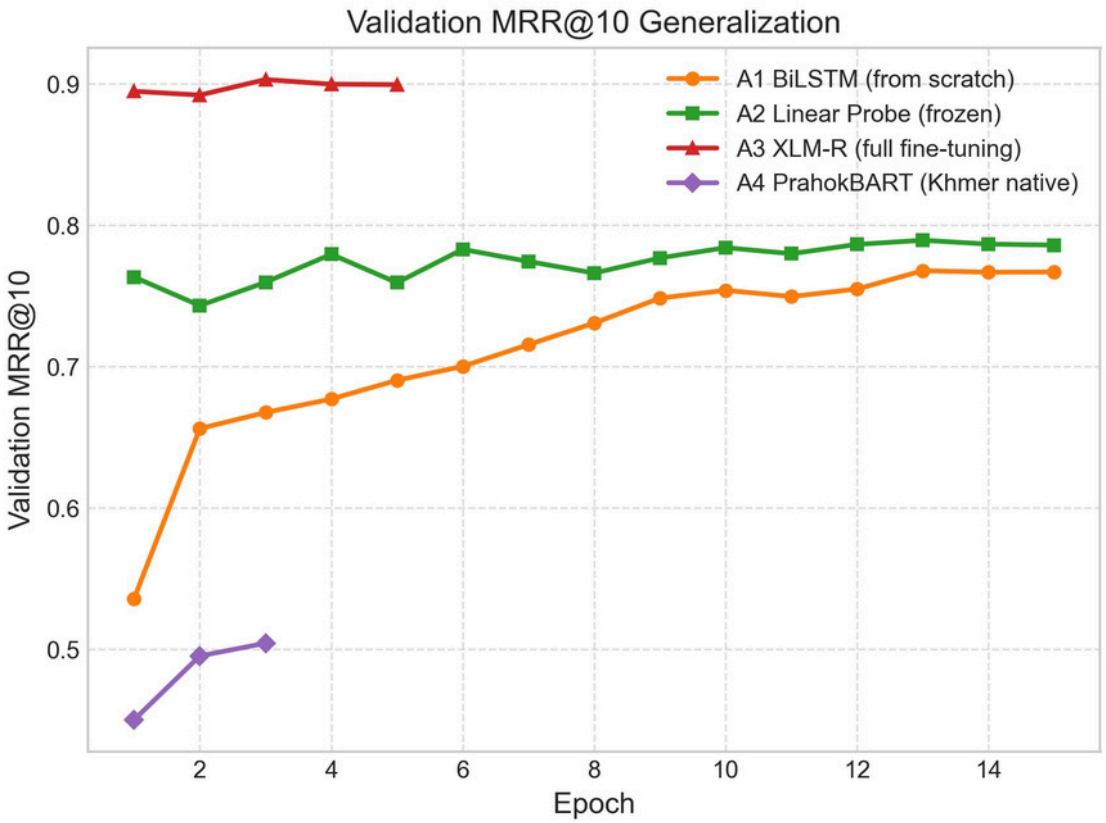
SELECTED CONFIGURATION

2e-5	0.01
learning rate	Weight decay

VAL MRR@10 **0.9095**

The learning curves told us which approaches were working

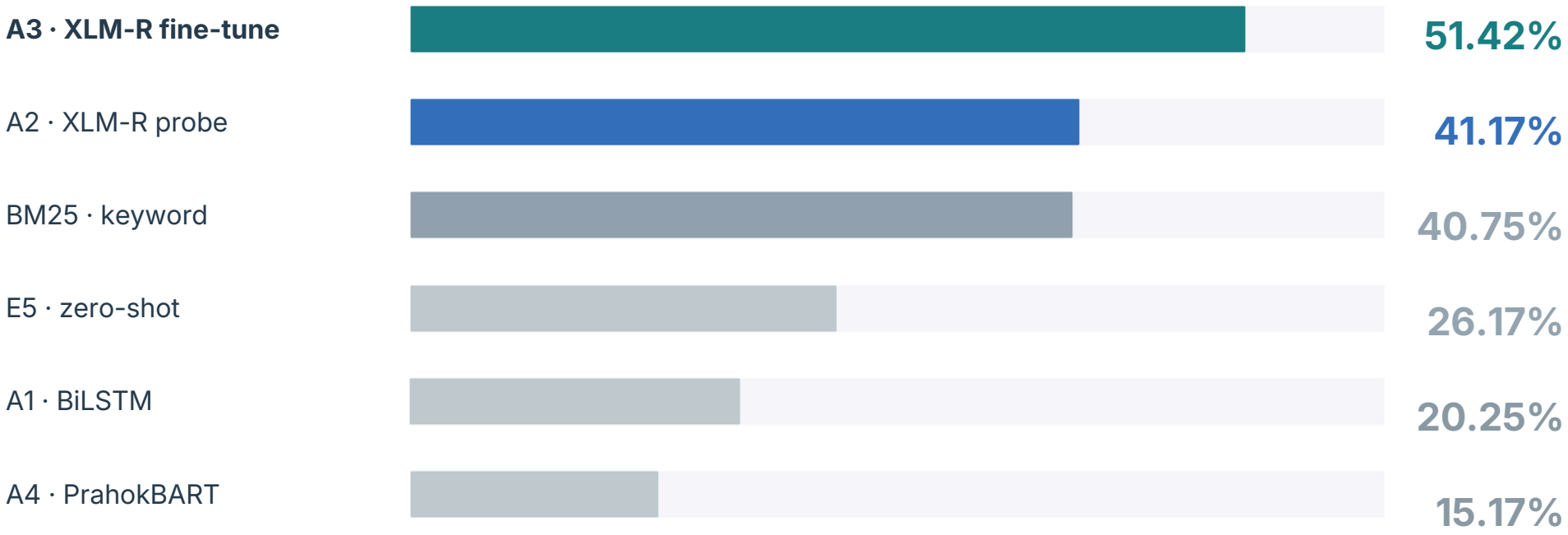




A3: 6 LR/weight-decay runs · selected LR 2×10^{-5}, WD 0.01 · validation MRR@10 0.9095 · Tesla T4

Full fine-tuning performed best on natural Khmer questions

RECALL@5 — SHARE OF RELEVANT ARTICLES IN TOP FIVE



A3 VS BM25

+10.67

pp Recall@5

+44.1%

relative MRR

MRR@10: A3 0.4827 · BM25 0.3350 | Recall@5 difference: 10.67 percentage points

EXPERIMENTAL EVIDENCE

Results on the same 200 Khmer questions

All 1,976 articles are candidates. No statute filter is applied in the benchmark.

System	Recall@1	Recall@5	MRR@10	Hit@5
BM25	0.2025	0.4075	0.3350	0.4700
E5 zero-shot	0.1625	0.2617	0.2588	0.3200
A1 BiLSTM	0.0875	0.2025	0.1724	0.2450
A2 XLM-R probe	0.2400	0.4117	0.3591	0.4750
A3 XLM-R fine-tune	**0.3375**	**0.5142**	**0.4827**	**0.5700**
A4 PrahokBART	0.0525	0.1517	0.1035	0.1800

Recall@5: mean fraction of relevant articles recovered; Hit@5: any relevant article found.

MRR@10: reciprocal rank of the first relevant article, averaged over questions.

Source: results/metrics/summary.csv. BM25 and zero-shot E5 are additional baselines.

The same model got 99% on titles, but 51% on real questions

TITLE-STYLE QUERIES

148 held-out titles

99.32%

NATURAL KHMER QUESTIONS

200 benchmark questions

51.42%

We trained mostly on legal titles, but people ask questions in very different ways.

Even the best model still makes important mistakes

60.5%

Vocabulary mismatch

The questionusesdifferent wording from the article.

19.8%

Near-misses

Therightarticleappeared at ranks 6–10.

17.4%

Multiple articles

Aquestionmayneedmore than one provision.

Limits: two codes, 200 questions, one seed; manual extraction error rate still unmeasured.

Can we retrieve the right article from 1,976 candidates?

BENCHMARK QUESTION · TQ-CIV-009

តើនរណាខ្លះអាចស្នើសុំឱ្យតុលាការប្រកាសបើកអាណាព្យាបាលទូទៅ?

REFERENCE ANSWER

Civil Code · Article 24

BM25

Keyword search

A3 · Fine-tuned XLM-R

Semantic search

Live check: same query · top 5 · all statutes · compare Article 24 ranking.

What I learned from comparing four neural retrievers

01 **Pretraining helped**

The from-scratch BiLSTM struggled on a small dataset.

02 **Fine-tuning won**

A3 was the strongest model on natural questions.

03 **Data still mattered**

99.32% title Recall@5 did not mean 99% on real questions.

Next: train with more natural Khmer questions and harder negative examples.

Model specifications

	A1 BiLSTM	A2 Probe	A3 Fine-tune	A4 BART
Pretraining	None	E5-base	E5-base	Khmer BART
Trainable params	1.97M	0.59M	278.04M	35.83M
Embedding size	512	768	768	512
Learning method	From scratch	Frozen backbone	All layers	Encoder fine-tune
Tokenization	Khmer words	SentencePiece	SentencePiece	Khmer subwords

How A3 was tuned

Sixlearning-rate/weight-decayconfigurations,selected by validation MRR@10.

VALIDATION MRR@10

	Weight decay 0.00	Weight decay 0.01
LR 1×10^{-5}	0.9023	0.9023
LR 2×10^{-5}	**0.9095**	**0.9095**
LR 3×10^{-5}	0.9082	0.9081

Selected: LR 2e-5, WD 0.01; tied at reported precision with WD 0.00. Best epoch 4 of 5.

Statistical comparison

Pairednon-parametricbootstraponthe200naturalbenchmark questions.

Comparison	Metric	Difference	95% CI	p-value
A3 vs BM25	Recall@5	**+0.1067**	[+0.0500, +0.1633]	0.0002
A3 vs BM25	MRR@10	**+0.1477**	[+0.0962, +0.1993]	0.0001
A2 vs BM25	Recall@5	**+0.0042**	[-0.0567, +0.0625]	0.9031
A3 vs A2	Recall@5	**+0.1025**	[+0.0492, +0.1575]	0.0001

These results support an improvement on this benchmark; they do not guarantee correct legal retrieval for every question.

Reproducing and inspecting the system

Data splits	src/dl/prepare_splits.py
Neural models	src/dl/models/
Contrastive objective	src/dl/losses.py
Training & tuning	src/dl/experiments/tune_a3.py
Evaluation & metrics	src/dl/evaluate.py · src/dl/metrics.py
Live Streamlit retriever	src/interfaces/ui/app.py

Demo prerequisite: A3 checkpoint and matching corpus embeddings loaded; never present BM25 fallback as A3.